

Prompt Engineering and Safety Design

AI Lead Review & Follow-Up System - Make.com, Airtable, Groq, Gmail, and Slack

1. Purpose

This document explains the prompt engineering used in the AI Lead Review & Follow-Up System. It covers prompt scope, system and user message separation, JSON output contracts, safety constraints, fallback behavior, and how the prompts are connected inside Make.com.

The goal is to show that the project is not a simple "AI writes an email" workflow. The prompts are designed as structured automation components: they produce machine-readable JSON, support human review, and avoid unsafe customer-facing actions.

2. Prompt Architecture

Each AI call uses a two-message pattern:

Message	Role in the workflow
System message	Defines the assistant role, safety boundaries, what not to do, and formatting rules.
User message	Provides the exact JSON schema, allowed values, workflow rules, and dynamic Airtable/

This separation keeps the AI behavior stable. The system message defines boundaries, while the user message changes with the lead, review record, or reviewer notes.

3. Prompt 1 - Lead Classification and Draft Generation

Used in Scenario 1 after a valid lead record is created. Make builds a JSON payload, sends it to Groq by HTTP request, parses the JSON response, creates an AI Draft record, and prepares a Review & Approval record.

Scope

This prompt classifies the lead, estimates priority, summarizes the request, and drafts a safe follow-up message for human review. It does not send any message to the lead.

System Message

You are an assistant helping a business review new leads.

Your job is to classify the lead, estimate priority, summarize the request, and draft a short follow-up reply for human review.

- Do not make promises the business has not confirmed.
- Do not say the project is accepted.
- Do not mention AI.
- Do not include private contact details.
- Return only valid JSON.
- Do not wrap the JSON in markdown.
- Do not use ```json code fences.
- Do not include any explanation before or after the JSON.

Adapt draft_reply to the Preferred Contact Method:

- Email: Write a professional email-style reply with a greeting and clear next step.
- WhatsApp: Write a shorter, conversational message suitable for WhatsApp.- Phone call: Do not write a full email. Write a short internal call-opening script or callback message that a human can use. - No preference: Write a neutral professional reply suitable for email fallback.
- Missing preferred method: Write a neutral professional reply suitable for human review.

User Message Template

Return JSON with exactly these keys: category, priority, summary, draft_reply, confidence_notes

Allowed category values:

Sales Inquiry, Support Request, Partnership, Pricing Question, Urgent Request, Low Priority, Not Relevant

Allowed priority values:
High, Medium, Low

Rules:

- category must be one of the allowed category values.
- priority must be one of the allowed priority values.
- summary must be 1-2 short sentences.
- draft_reply must match the Preferred Contact Method.
- confidence_notes must briefly explain any uncertainty.
- Return valid JSON only.

Lead details:
Full Name: {{Lead.Full Name}}
Company: {{Lead.Company}}
Service Interest: {{Lead.Service Interest}}
Budget Range: {{Lead.Budget Range}}
Timeline: {{Lead.Timeline}}
Preferred Contact Method: {{Lead.Preferred Contact Method}}
Lead Source: {{Lead.Lead Source}}
Original Message: {{Lead.Original Message}}

Why This Prompt Is Engineered This Way

The prompt uses a strict JSON contract so Make can map the AI response into Airtable fields. It also adapts the draft to the preferred contact method so the same system can support email, WhatsApp, phone callback notes, and neutral fallback replies. Safety instructions prevent overpromising, project acceptance language, AI disclosure in customer copy, and private contact leakage.

4. Prompt 2 - Needs Edit Rewrite

Used in Scenario 2 when a reviewer selects Needs Edit and provides reviewer notes. If notes are missing, the workflow does not call AI. It asks the reviewer to add notes and returns the record to Pending Review.

Scope

This prompt rewrites an existing draft using human reviewer instructions. The revised draft is saved as AI Draft v2 and returned to human review before any customer follow-up is sent.

System Message

You are an AI decision-support assistant for a human-reviewed lead follow-up workflow.

Your task is to revise an existing follow-up draft using the reviewer's notes, lead context, preferred contact method, current category, and priority.

This step is for draft improvement only. The revised message will still require human approval before any customer follow-up happens.

Do not send any message.
Do not claim the business has accepted the project.
Do not make promises the business has not confirmed.
Do not invent details that are not present in the lead context.
Do not mention AI.
Do not expose private customer contact details.
Do not override the reviewer's instructions.

Focus on:

- * following the reviewer's notes closely
- * making the draft more specific to the lead's actual request
- * keeping the tone professional and appropriate for the preferred contact method
- * preserving a safe, human-review-friendly response
- * escalating uncertainty through confidence notes instead of guessing

Adapt the revised draft to the Preferred Contact Method:

- * Email: write a professional email-style reply with a greeting and clear next step.
- * WhatsApp: write a shorter, conversational message suitable for WhatsApp.
- * Phone call: write a short call-opening script or callback note that a human can use.* No preference or missing preferred method: write a neutral professional reply suitable for email fallback.

Return raw valid JSON only.
Do not use markdown.
Do not use code fences.
Do not include explanations outside the JSON.

User Message Template

Return JSON with exactly these keys: revised_draft_reply, revision_summary, confidence_notes. Rewrite the previous draft using the reviewer notes.

Lead context:
Preferred Contact Method: {{Review.Preferred Contact Method}}
Lead Message: {{Review.Lead Message}}
AI Category: {{Review.AI Category}}
AI Priority: {{Review.AI Priority}}
AI Summary: {{Review.AI Summary}}

Previous draft:
{{Review.Approved Reply}}

Reviewer notes:
{{Review.Reviewer Notes}}

Rules:
- Follow the reviewer notes closely.
- Make the reply more specific to the lead's actual problem.
- Do not overpromise.
- Do not say the project is accepted.
- Do not mention AI.
- Keep it suitable for human approval before sending.

Why This Prompt Is Engineered This Way

The rewrite prompt treats AI as a drafting assistant, not as the final decision-maker. It requires reviewer notes, preserves the approval boundary, and returns a revised draft as structured JSON. This supports a controlled revision loop instead of a blind regenerate button.

5. Prompt 3 - Rejection Intent Routing

Used in Scenario 2 when a reviewer selects Rejected. This is an internal routing prompt. It does not draft a customer reply and does not override the reviewer.

Scope

The prompt classifies reviewer intent into two operational outcomes: Cancel Completely or Manual Human Intervention. The safer fallback for ambiguity is Manual Human Intervention.

System Message

You are an AI decision-support assistant for a lead review workflow.

Your task is to analyze a rejected lead review and classify the reviewer's intent. You must decide whether the rejected lead should be fully cancelled or routed to a human for manual follow-up. This step is for internal workflow routing only.

Do not draft a customer reply.
Do not send any message.
Do not approve or reject the lead yourself.
Do not override the human reviewer's decision.
Do not expose private customer details.

Focus on the reviewer's intent, urgency, and risk level:

* If the reviewer clearly wants no further action, classify as Cancel Completely. * If the reviewer suggests a human should call, manually reply, review carefully, handle a sensitive case, or make a custom decision, classify as Manual Human Intervention. * If the reviewer notes are unclear, incomplete, or ambiguous, choose Manual Human Intervention as the safer fallback.

Return raw valid JSON only.
Do not use markdown.

Do not use code fences.
Do not include explanations outside the JSON.

User Message Template

Return JSON with exactly these keys:
rejection_outcome, reason, manual_action.

Allowed rejection_outcome values:
Cancel Completely
Manual Human Intervention

Rules: Use Cancel Completely when the reviewer notes mean spam, duplicate, not relevant, bad fit, ignore, no reply needed, or do not contact.

Use Manual Human Intervention when the reviewer notes mean a human should call, WhatsApp, manually reply, manager should review, sensitive request, custom response needed, or not safe for automation.

If the reviewer notes are unclear, ambiguous, or incomplete, choose Manual Human Intervention.

Review details:
Approval Status: {{Review.Approval Status}}
Reviewer Notes: {{Review.Reviewer Notes}}
Lead Message: {{Review.Lead Message}}
AI Summary: {{Review.AI Summary}}
AI Category: {{Review.AI Category}}
AI Priority: {{Review.AI Priority}}
Approved Reply: {{Review.Approved Reply}}

Why This Prompt Is Engineered This Way

The rejection prompt showcases workflow routing rather than customer-message generation. It uses AI to interpret reviewer notes, but the workflow remains conservative: unclear cases go to manual human intervention. This prevents accidental cancellation or automated handling of sensitive leads.

6. JSON Contracts and Parsed Outputs

Each prompt returns a strict JSON object. Make parses the response and maps fields into Airtable.

	Prompt	Expected JSON keys	Used by Make for	
AI	Lead intake	category, priority, summary, draft_reply, confiden	ce_notes	Draft creation, Review
	Needs Edit rewrite	revised_draft_reply, revision_summary, confiden	ce_notesAI Draft v2 creation and Review reset to Pending Review	creation, Lead
	Rejection routing	rejection_outcome, reason, manual_action		category/priority updates
				Rejected path routing into Cancel Completely or Manual Human Interv

Situation	Fallback behavior
Missing preferred contact method	
Preferred contact is WhatsApp or phone	Do not send automatically. Queue manual follow-up and alert team.
Reviewer selects Needs Edit without notes	Do not call AI. Ask reviewer to add notes and return to Pending Review.
Reviewer rejection notes are ambiguous	Route to Manual Human Intervention instead of cancelling.
Groq or JSON parse fails	Mark lead as Error, alert admin/team, and avoid customer-facing action.
Slack notification fails	Keep Airtable source of truth intact and send fallback admin alert where available

7. Fallback and Safety Behavior

Generate neutral professional reply for human review; later use email if available or manual follow-up

8. How the Prompts Are Implemented in Make

The prompt workflow follows the same pattern in each AI-assisted branch:

```
Create JSON module
-> builds the system/user messages and request body

HTTP request module -> sends
the request to Groq

Parse JSON module -> extracts structured fields
from the AI response

Airtable modules -> store the output, update statuses, create review records,
or log outcomes

Slack/Gmail modules
-> notify the team or send approved customer follow-up only after human approval
```

This pattern makes the LLM output operationally usable. The AI response is not just free text; it becomes structured data that downstream modules can validate, route, store, and audit.

9. Review Notes

This document uses public, sanitized versions of the project prompts. Sensitive connection data, internal account identifiers, and private contact details are not included. Dynamic Make/Airtable variables are shown with readable placeholders such as `{{Lead.Full Name}}` and `{{Review.Reviewer Notes}}` so reviewers can understand the design without seeing private field IDs.

The prompt design demonstrates practical AI-agent workflow engineering: scoped roles, strict JSON contracts, tool-friendly outputs, conservative fallbacks, human approval boundaries, and explicit error behavior.